

Secure Coding using C/C++	
Course Code: CYE551	Credits: 3:0:0
Prerequisite: Nil	Contact Hours: 42L
Course Coordinator: Mrs. Shankaramma	

Course Contents

Unit I

Running with Scissors: Gauging the Threat, Security Concepts, Development Platforms, **Strings:** Character Strings, Common String Manipulation Errors, String Vulnerabilities and Exploits, Mitigation Strategies for Strings, String- Handling Functions.

- Pedagogy: Chalk and Talk, PowerPoint Presentations

Unit II

Pointer Subterfuge: Data Locations, Function Pointers, Object Pointers, Modifying the Instruction Pointer, Global Offset Table, The .dtors Section, Virtual Pointers, The atexit() and on_exit() Functions, The longjmp() Function, Exception Handling. **Dynamic Memory Management:** C Memory Management, Common C Memory Management Errors, C++ Dynamic Memory Management, Common C++ Memory Management Errors.

- Pedagogy: Chalk and Talk, PowerPoint Presentations

Unit III

Memory Managers, Doug Lea's Memory Allocator, Buffer Overflows on the Heap, Notable Vulnerabilities. **Integer Security:** Introduction to Integer Security, Integer Data Types, Integer Conversions, Integer Operations.

- Pedagogy: Chalk and Talk, PowerPoint Presentations

Unit IV

Integer Vulnerabilities, Mitigation Strategies. **Formatted Output:** Variadic Functions, Formatted Output Functions, Exploiting Formatted Output Functions, Stack Randomization, Notable Vulnerabilities.

- Pedagogy: Chalk and Talk, PowerPoint Presentations

Unit V

File I/O: File I/O Basics, File I/O Interfaces, Access Control, File Identification, Race Conditions, **Recommended Practices:** The Security Development Lifecycle, Security Training.

- Pedagogy: Chalk and Talk, PowerPoint Presentation

Text Books:

1. Robert C. Seacord: Secure Coding in C and C++, 2nd Edition, Pearson, 2013. (Chapter 1,2,3,4,5,6,8,9).

References:

1. SEI CERT Coding Standard

Course Outcomes (COs):

At the end of the course, students will be able to

1. Understand the vulnerability associated with Sting Handling and Mitigation strategies. (PO1,2,3,PSO-1,2)
2. Describe the threats associated with Runtime Memory Management and Pointers. (PO1,2,3,PSO-1,2)
3. Apply the mitigation techniques for vulnerable memory management and data type conversion. (PO1,2,3,4,5,PSO-1,2)
4. Identify the vulnerability associated with Formatted Output and Concurrency (PO1,2,3,PSO-1,2)
5. Describe the vulnerability mitigation techniques for File I/O and recommended practices for security development lifecycle. (PO1,2,3,8,PSO-1,2,3)

Course Assessment and Evaluation:

Continuous Internal Evaluation (CIE): 50 Marks		
Assessment Tools	Marks	Course Outcomes (COs) addressed
Internal Test-I (CIE-I)	30	CO1, CO2
Internal Test-II CIE-II)	30	CO3, CO4, CO5
Average of the two CIE shall be taken for 30 marks		
Other Components		
Case study	10	CO1, CO2, CO3,CO4, CO5
Programming Assignment	10	CO1, CO2, CO3,CO4, CO5
The Final CIE out of 50 Marks = Average of two CIE tests for 30 Marks+ Marks scored in Case Study +Marks scored in Assignment		
Semester End Examination (SEE)	50	CO1, CO2, CO3 CO4, CO5